Week 8 Workshop Information Needs and Training Data Generation

IFN647

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Notification

Assignment 2:

- Assignment 2 will be released in Week 8 and will be a group assignment.
- You are required to form a group of three students from Week 7 to Week 8.
- Please ensure that all group members are enrolled in the same workshop. Then, select a group on Canvas under Group/Assignmnet 2 and complete the self sign-up by the end of Week 8 (26 April 2026).
- **On Monday (27 April 2026), any students who are not yet in a group will be randomly allocated**.
- Please note that no changes to groups will be permitted after this date.

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Guidelines for Group Formation and Deliverables

- Limit group size to a maximum of three members.
- **Leadership:** Canvas will automatically assign a group leader (the first student to join the group will be designated as the leader).
- **Deliverables**
  - **Demonstration (Week 13 Workshops):**
    The demo must be completed prior to the final report submission deadline. Each group is required to demonstrate only their custom model (*Model_C*), including:
    - **Design:** the underlying algorithm
    - **Implementation:** a working/running system
    - **Evaluation:** a table reporting the performance of *Model_C* on the three required metrics
    - **Individual contributions:** specified by task or expressed as a percentage.
  - **Final Report:** Due on **3 June 2026**.
    - **Submission Requirements:**
      - Only one submission per group is required.
      - The submission must include both the final report and the complete source code.

Workshop Objectives

1. Understand how to capture user information needs as the underlying drivers of queries.
2. Understand how to design an algorithm based on the BM25 IF model.
3. Generate training data using the Pseudo-Relevance Hypothesis (i.e., “top-ranked documents are assumed to be relevant”) and evaluate the resulting performance.

Workshop Task 1

"Pseudocode" is a method of representing algorithms using a mixture of natural language and programming language-like syntax. It is used to outline the logic of a program

- Design a BM25 based IR model which uses query Q to rank documents in the "Training_set" folder and save the result into a text file; e.g., BaselineModel_R102.dat; where each row includes the document number and the corresponding BM25 score or ranking (in descendent order).
- Formally describe your idea in an algorithm → **Using Pseudo Code**
- **PS:** We believe everyone in the class should know how to do <u>text pre-processing</u> and calculate BM25 scores; therefore, we don't need to discuss the details of these parts when you describe your ideas in an algorithm.

STEPS:
- Read data
- Get Document ID
- Find content in <text></text>
- Get tokens(terms) in each document and their term frequencies
- Get each term's document frequency

- Input:
  Q - Query
  Training_set – set of .xml documents

- Output
  BaselineModel_R102.dat

73038	5.898798484774149
26061	4.273638903483098
65414	4.1414522450167475
57914	3.967136888209526
58476	3.708467957856744
76635	3.5867337114200843
12769	3.4341129093591456
12767	3.352170358051889

Workshop Task 1 Knowledge Extension

Simple pseudo code template

```
Algorithm [name of the algorithm]
Inputs: [list of inputs and their types]
Outputs: [list of outputs and their types]

[Main algorithm steps]
Step 1: [description of step 1]
    [code for step 1]
Step 2: [description of step 2]
    [code for step 2]
    ...
Step n: [description of step n]
    [code for step n]
[Optional: define subroutines or functions that the main algorithm uses]
End Algorithm
```

Workshop Task 2

Task 2. Training set generation

Pseudo-Relevance Hypothesis: “top-ranked documents (e.g., documents’ BM25 scores are great than 1.00) are possible relevant”.

Design a python program to find a training set $\underline{D}$ which includes both $\underline{D}^+$ (positive – likely relevant documents) and $\underline{D}^-$ (negative – likely irrelevant documents) in the given un-labelled document set U.

The input file is “BaselineModel_R102.dat” or BaselineModel_R102_v2.dat, and The output file name is “PTraining_benchmark.txt”, which has the following sample output:

```
R102 73038 1
R102 26061 1
R102 65414 1
R102 57914 1
R102 58476 1
R102 76635 1
R102 12769 1
R102 12767 1
R102 25096 1
R102 78836 1
…
R102 86912 0
R102 86929 0
R102 11922 0
R102 14713 0
R102 11979 0
…
```

$\underline{D}^+$ Positive – likely relevant documents

$\underline{D}^-$ Negative – likely irrelevant documents

Output File: The labelled document set D

PTraining_benchmark.txt

Input File: The unlabelled document set U

BaselineModel_R102.dat

```
73038 5.898798484774149
26061 4.273638903483098
65414 4.141452245016747 5
57914 3.967136888209526
58476 3.708467957856744
76635 3.5867337114200843
12769 3.434112909359145 6
12767 3.352170358051889
```

Workshop Task 2 cont.

Note:

Please note that this week we do not require you to use BM25 to generate rankings.

We used a python program to generate "BaselineModel_R102.dat" using the original BM25 equation (where $R=r(t)=0$), but it contains some negative BM25 values because $N < (2n(t) + 1)$ for some term t.

To simple fix the negatives, we just replaced N with 3N (you could use the method mentioned in week 7 lecture notes for the negative score issue in BM25), then we can generate "BaselineModel_R102_v2.dat".

Workshop Task 3

Task 3. Discuss the precision and recall of the pseudo-relevance assumption.

Compare the relevance judgements (the true benchmark – "Training_benchmark.txt") and the pseudo one ("PTraining_benchmark.txt") and display the number of relevant documents, the number of retrieved documents, the recall, precision and F1 measure by using pseudo-relevance hypothesis, where we view documents in $\underline{D}^{+}$ as retrieved documents). You may use the function you designed in week 7). The following are possible outputs:

```
the number of relevant docs: 106
the number of retrieved docs: 25
the number of retrieved docs that are relevant: 20
recall = 0.18867924528301888
precision = 0.8

F-Measure = 0.30534351145038174
```

Please analyse the above outputs to understand possible uncertainties (or errors) in the generated training set "PTraining_benchmark.txt". You can also discuss any possible ways to generate a high-quality training set.

Workshop Task 3 cont.

Relevant / Not relevant
Actual

Retrieved / Not retrieved
Given by pseudo relevance hypothesis

Training_benchmark.txt

```
R102 10182 0
R102 11083 0
R102 11481 1
R102 11485 1
R102 11922 1
R102 11923 1
R102 11930 1
R102 11960 0
R102 11979 1
R102 12479 1
```

PTraining_benchmark.txt

```
R102 73038 1
R102 26061 1
R102 65414 1
R102 57914 1
R102 58476 1
R102 76635 1
R102 12769 1
R102 12767 1
R102 25096 1
R102 78836 1
R102 82227 1
R102 26611 1
R102 24515 1
R102 33172 1
```